

Classifiers as Modifiers: A Stranding Analysis of Korean Numeral Constructions

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1 Introduction

There are mainly three types¹ of numeral classifier constructions in Korean:

- (1) ‘Two children bought a book.’

a. **Noun Initial (NI) Type:**

ai-tul twu myeong-i chaek-ul sa-ess-ta.
child-PL two CL-NOM book-ACC buy-PST-DECL

b. **Floated Numeral Classifier / Quantifier (FQ) Type:**

ai-tul-i twu myeong(-i) chaek-ul sa-ess-ta.
child-PL-NOM two CL(-NOM) book-ACC buy-PST-DECL

c. **Genitive Case (GC) Type:**

twu myeong-ui ai-tul-i chaek-ul sa-ess-ta.
two CL-GEN child-PL-NOM book-ACC buy-PST-DECL

The main syntactic difference can be found by the position of the nominative case marker *-i*. In the Noun Initial (NI) type (1a), the case marker appears after the NUM-CL *twu myeong*. In the Floated Quantifier (FQ) type (1b), the marker appears between the main DP *ai-tul* and NUM-CL with an optional nominative case marker after NUM-CL. In the Genitive Case (GC) type (1c), NUM-CL is fronted with a genitive case marker *-ui* and the nominative case marker appears after the main DP. These constructions can appear similarly in object positions by using an accusative case marker *-ul*. In addition to this syntactic difference, Kim (2005) claims that there exists a semantic difference on the availability of specific or non-specific reading of the main DP; however, semantic aspects of these constructions are out of scope of this paper.

While the internal structure of DPs with numeral and classifier is unsettled, many approaches root their analysis on the assumption that the agreement between NP and classifier is syntactic (Li, 1999; Watanabe, 2006; Park, 2009; Choi, 2011; Park, 2022). Regardless, a syntactic proximity between the noun, numeral, and classifier remains uncontroversial. However, as Kim (2013) and others mentioned, adverbial elements can surface between the head DP and NUM-CL (henceforth CIP), as in (2).

- (2) **Adverbs surfacing between head DP and CIP:**

- a. ai-tul-i eojae twu myeong(-i) chaek-ul sa-ess-ta.
child-PL-NOM yesterday two CL(-NOM) book-ACC buy-PST-DECL
‘Two children bought a book yesterday.’

- b. ai-tul-i jeongmal twu myeong(-i) chaek-ul sa-ess-ta.
child-PL-NOM really two CL(-NOM) book-ACC buy-PST-DECL
‘Two children really bought a book.’

(modified from Kim, 2013)

¹Following Kim (2013)’s abbreviation

To account for the intervention of adverbial elements, I propose that it is a result of the stranding of CIP, which extends the analysis of CIP as a modifier of the head DP. I first examine the morphosyntactic properties of numeral classifiers in Korean and motivate the modifier analysis to derive various classifier constructions (Section 2). Then, I extend this analysis to the distributive marker *-ssik*, demonstrating that DistP exhibits parallel stranding behavior (Section 3). I also investigate complex distributive constructions (stacked modifiers), providing further structural evidence for the adjunct status of CIP and DistP (Section 4). I argue that this evidence suggests an inherent feature of CIP and DistP ([$\pm$ cardinality]) to account for an ungrammatical configuration within complex distributive constructions.

2 The Syntax of Numeral Classifiers

2.1 Morphosyntactic Properties

Korean is a classifier language that requires classifier when counting the numbers of the noun. The classifier should agree with the head noun, as in (3).

(3) **Feature agreement between noun and classifier:**

- | | |
|--|--|
| <p>a. ai-tul sey myeong
child-PL three CL_{human}
‘three children’</p> | <p>b. goyang’i-tul daseot mari
cat-PL five CL_{animal}
‘five cats’</p> |
|--|--|

Native Korean numerals² must appear in their attributive form before classifiers³, forming a strictly adjacent unit. While Lee (2000) and Park (2022) argue that the plural marker *-tul* and CIP cannot co-occur (except with human nouns), I follow Kim (2009) in assuming that classifiers do not inherently distinguish between singular and plural. Instead, their sole function is counting. This assumption is critical in analyzing CIP as a modifier of the head DP (nP + PL), which I develop further in Section 2.3.

2.2 Intervention of Adverbs in DP

Among three types of numeral classifier constructions in Korean, only the FQ type allows surfacing of an adverb between the head DP and CIP. Not all adverbs can surface in this position. Temporal adverbs (e.g., *eojae* ‘yesterday’, *gakkeum* ‘occasionally’) or modal adverbs (e.g., *jeongmal* ‘really’, *amado* ‘maybe’) can while manner adverbs (e.g., *ppalli* ‘quickly’) cannot, as in (4)⁴.

(4) **Adverbs surfacing between DP and CLP in FQ type:**⁵

- | | |
|---|--|
| <p>a. ai-tul-i gakkeum twu myeong(-i) chaek-ul sa-n-ta.
child-PL-NOM occasionally two CL(-NOM) book-ACC buy-NPST-DECL
‘Two children occasionally buy a book.’</p> | <p>b. ai-tul-i amado twu myeong(-i) chaek-ul sa-ess-ta.
child-PL-NOM maybe two CL(-NOM) book-ACC buy-NPST-DECL
‘Two children maybe bought a book.’</p> |
|---|--|

²Native Korean numerals have distinct attributive forms for 1 to 4 (presented here as *base* vs. *attributive*): **1** (*hana* vs. *han*), **2** (*twul* vs. *twu*), **3** (*set* vs. *se*), and **4** (*net* vs. *ne*). From **5** onwards (*daseot*, *yeoseot*, etc.), the base and attributive forms are generally identical. **20** is an exception, having *sumul* as a base form but *sumu* in its attributive form.

³Human nouns are exception to the numeral-classifier construction. That is, #P is not required to merge with CIP to modify the head DP. Despite this exception, the main syntactic phenomenon of CIP stranding remains.

⁴Tense in (4a) was modified to non-past for semantic agreement with ‘occasionally’.

⁵CIPs (in this paper) are in bold to help locating it.

- c. *ai-tul-i ppalli **twu myeong**(-i) chaek-ul sa-ess-ta.
 child-PL-NOM quickly **two** CL(-NOM) book-ACC buy-NPST-DECL
 Intended: ‘Two children quickly bought a book.’

In contrast, no adverbs can surface between the head DP and CIP in the NI and GC types:

(5) **No adverbs can surface between DP and CIP in NI and GC types:**

- a. *ai-tul eojae **twu myeong**-i chaek-ul sa-ess-ta.
 child-PL yesterday **two** CL-NOM book-ACC buy-PST-DECL
 Intended: ‘Two children bought a book yesterday.’
- b. ***twu myeong**-ui eojae ai-tul-i chaek-ul sa-ess-ta.
two CL-GEN yesterday child-PL-NOM book-ACC buy-PST-DECL
 Intended: ‘Two children bought a book yesterday.’

Regarding the optionality of nominative case marking on CIP in the FQ construction, Schütze (2001) argues that such NOM marking is its secondary function as a focus marker. He proposes the generalization that “a genuine case on a head noun licenses the presence of the same case on an associated floated quantifier” (p. 9). To account for the syntactic mechanism of the case assignment, I follow Levin (2017)’s argument that the Korean case system supports the Dependent Case model (Marantz, 1991).

2.3 Structural Proposal: CIP as Modifier

Although agreement is required between a head noun and a classifier, I argue that this relation is not purely syntactic. Specifically, under certain conditions, the violation of canonical agreement is possible. As shown in (6a), a classifier for animals may be used with a human noun to convey a pejorative stance (e.g., highlighting animal-like behavior). Likewise, in (6b), an honorific classifier for humans of high status may be used with a non-human noun to achieve a glorifying or anthropomorphic effect.

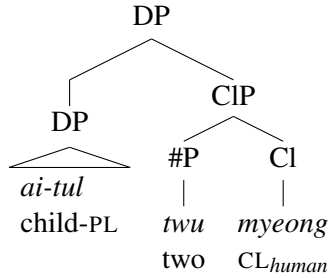
(6) **Grammatical DP despite feature mismatch between N and Cl:**

- | | |
|---|--|
| <p>a. saram han mari
 person one CL_{animal}
 ‘one person’</p> | <p>b. goyang’i han bun
 cat one CL_{human.HON}
 ‘one cat’</p> |
|---|--|

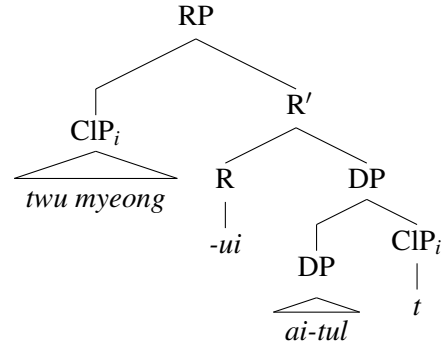
Given that the function of a classifier is counting, and that agreement violation between a head noun and a classifier is possible, I propose that CIP functions as a modifier of a DP, as in (7a). This analysis suggests that the NI type is a base generated structure of the numeral classifier construction. The difference between the NI type and the FQ type is not identifiable yet. The GC type is an instance of DP-internal predicate inversion (Den Dikken, 2006), as in (7b). A functional particle (which is also a genitive marker in Korean) *-ui* is a head of Relator Phrase (RP) that functions similar to English *of*.

(7) **Internal Structure of DPs with CIP:**

a. **NI and FQ types:**



b. **GC type:**



2.4 CIP Stranding

2.4.1 CIP Stranding in NI and FQ Type

I propose that the grammaticality of (2a) (repeated here as (8b)) is a result of CIP stranding. That is, the NI type (8a) is a consequence of the movement of the entire complex DP to Spec,TP, as in (8a). However, in the FQ type (8b), only the inner DP moved to Spec,TP.

(8) ‘Two children bought a book yesterday.’

a. **NI Type with Temporal Adverb:**

ai-tul **twu myeong**-i eojae chaek-ul sa-ess-ta.
child-PL **two** CL-NOM yesterday book-ACC buy-PST-DECL

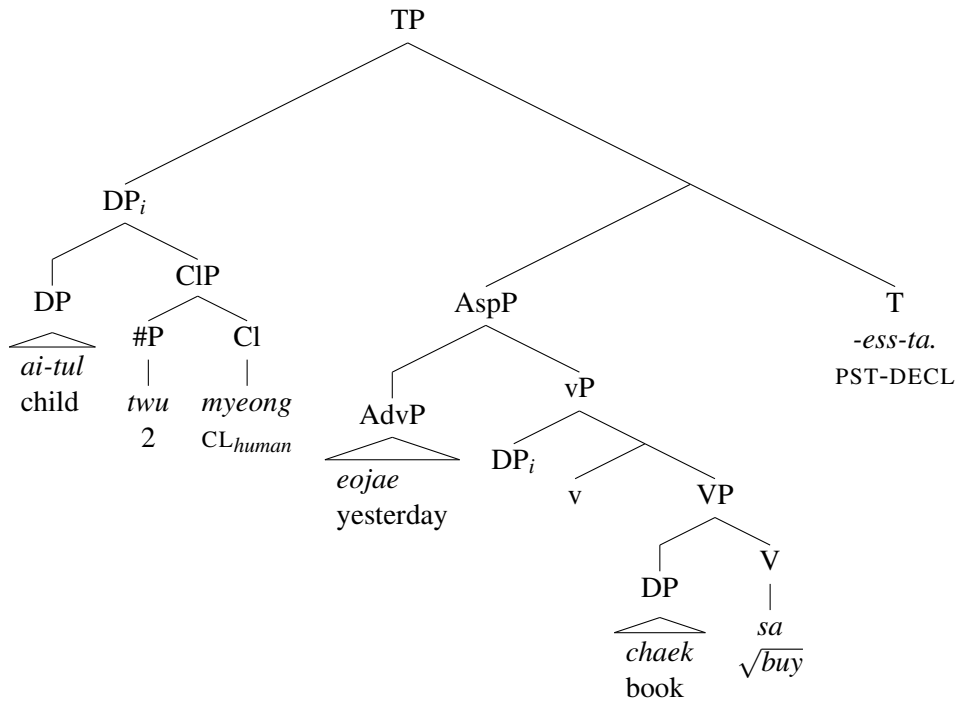
b. **FQ Type with Temporal Adverb (CIP stranding):**

ai-tul-i eojae **twu myeong**(-i) chaek-ul sa-ess-ta.
child-PL-NOM yesterday **two** CL(-NOM) book-ACC buy-PST-DECL

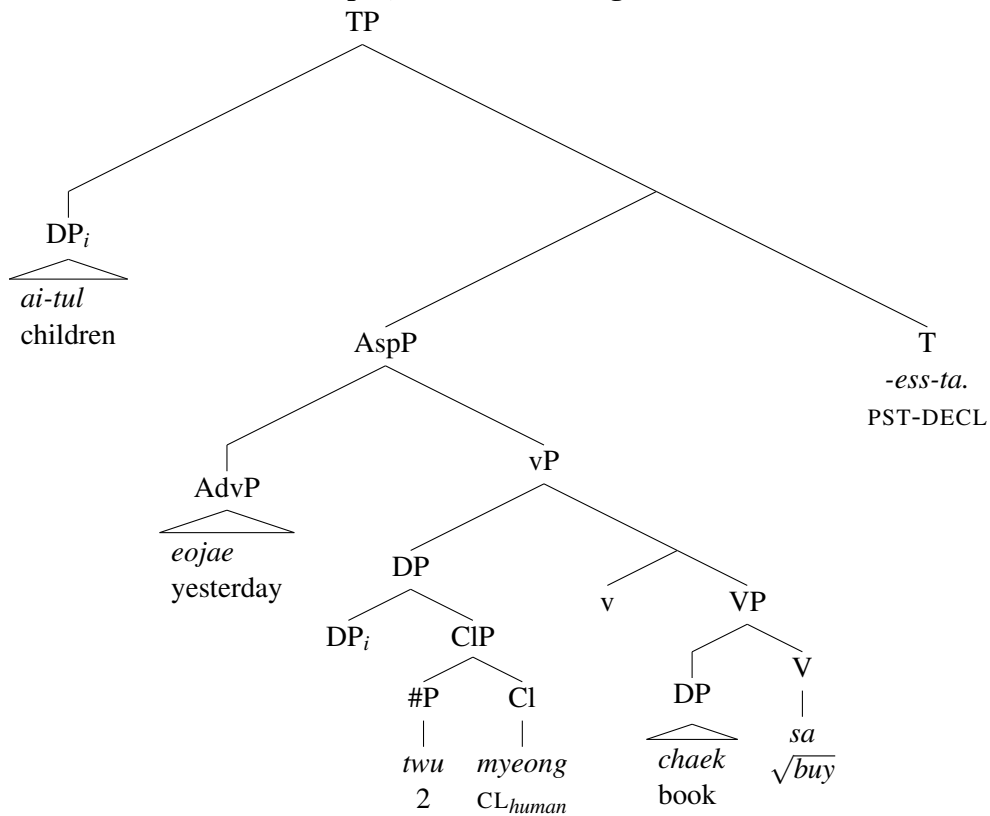
In (9), the movement of the entire complex DP *ai-tul twu myeong* to Spec,TP results from T searching for a DP to satisfy its EPP feature. This is a standard case of pied-piping (Ross, 1967), in which the CIP moves along with the head DP. After the movement, case assignment happens via the Dependent Case model: the complex DP receives NOM case and the lower DP *chaek* ‘book’ receives ACC case.

In (10), however, only the inner DP *ai-tul* moves to Spec,TP to satisfy the EPP feature. This results in CIP stranding where CIP is left behind in its base-generated position. The availability of this derivation supports the structural analysis of CIP as a modifier, which permits the extraction of the inner DP from the complex DP without violating phase constraints. Following the movement, case assignment happens. The inner DP *ai-tul* ‘children’ receives NOM case and the lower DP *chaek* ‘book’ receives ACC case. As discussed in Section 2.2, I assume that an optional case marking on CIP is not structurally licensed.

(9) Movement of complex DP to Spec,TP:



(10) Movement of inner DP to Spec,TP (CIP stranding):



2.4.2 CIP Stranding in GC Type

In contrast, CIP stranding is not possible in the GC type, which is a result of DP-internal predicate inversion. In (11a), the entire complex DP *twu myeong-ui ai-tul* moved to Spec,TP to satisfy the EPP feature. However, unlike (8b), CIP stranding by only having the movement of the inner DP to Spec,TP is no longer possible, as in (11b).

(11) ‘Two children bought a book yesterday.’

a. **GC Type with Temporal Adverb (Movement of complex DP is possible):**

twu myeong-ui ai-tul-i eojae chaek-ul sa-ess-ta.
two CL-GEN child-PL-NOM yesterday book-ACC buy-PST-DECL

b. **GC Type with Temporal Adverb (CIP stranding is not possible):**

*ai-tul-i eojae **twu myeong-ui** chaek-ul sa-ess-ta.
child-PL-NOM yesterday **two** CL-GEN book-ACC buy-PST-DECL

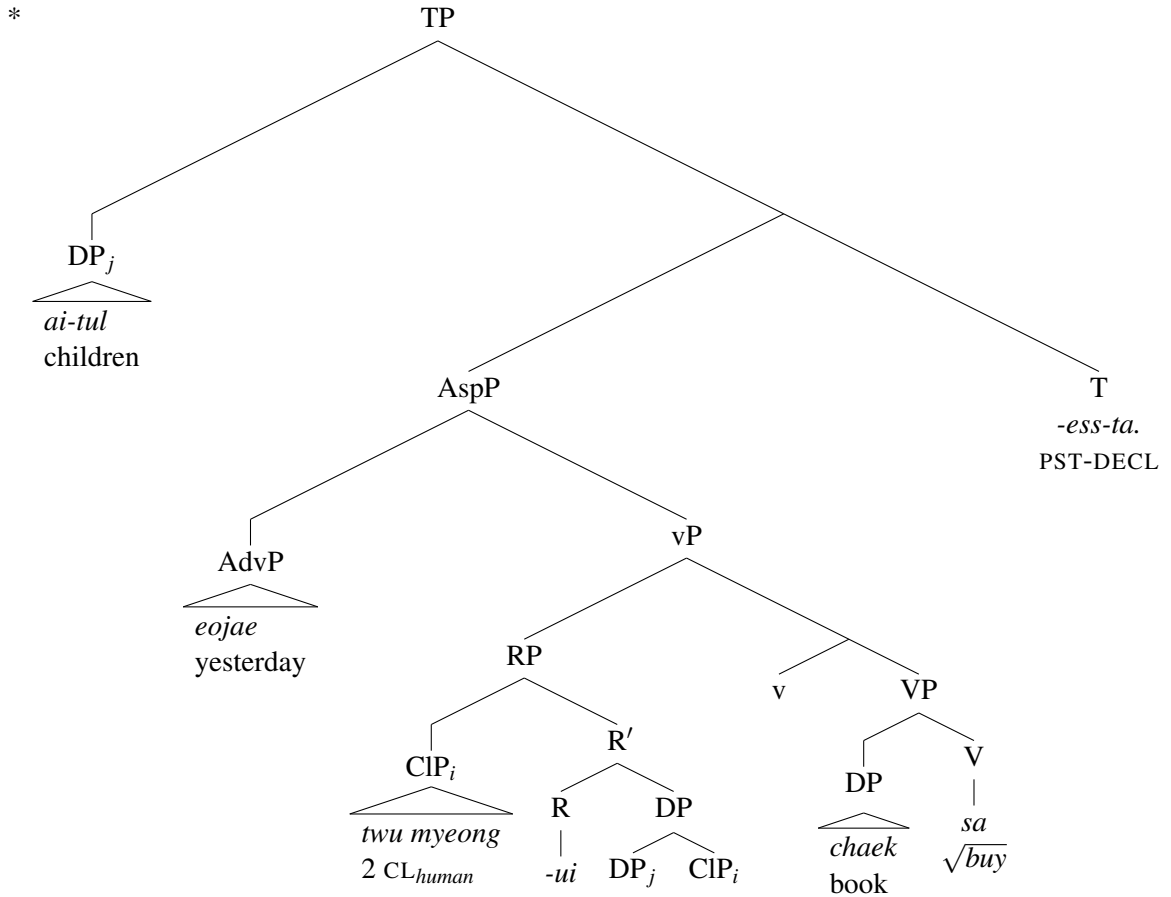
The ungrammaticality of extracting the inner DP from the Relator Phrase (RP) in (11b) can be accounted by the Phase Impenetrability Condition (PIC) (Chomsky, 2001). I follow Den Dikken (2006)’s claim that RP is a phase.

Phase Impenetrability Condition (PIC):

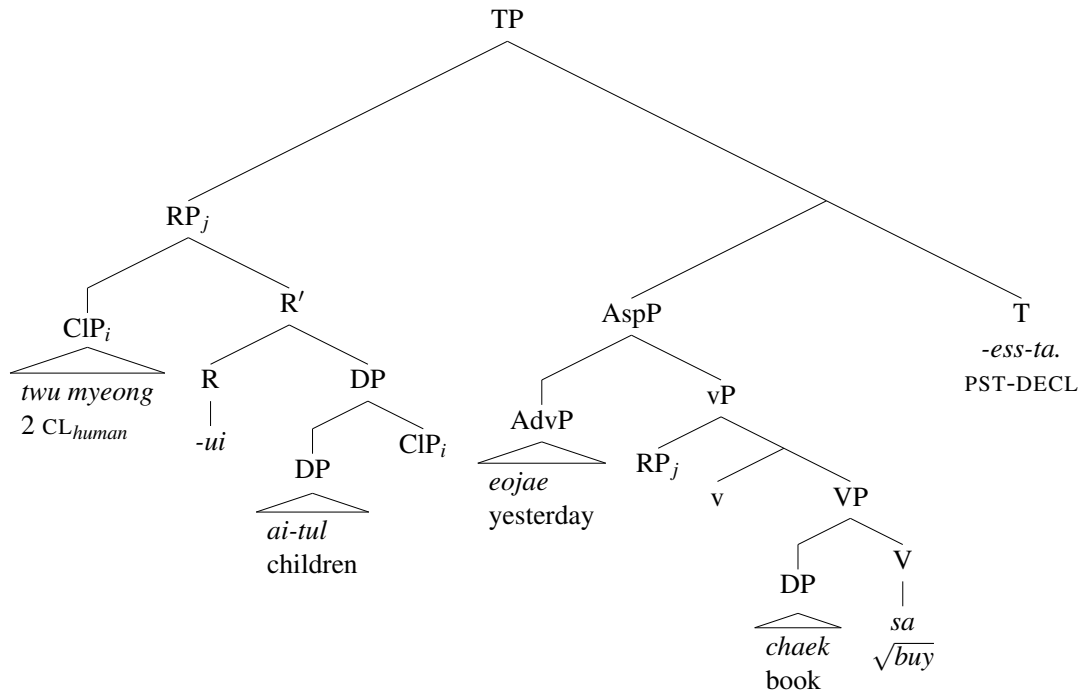
If Ψ is a phase with head H , then the domain of H is not accessible for operations outside Ψ .
Only H and its *edge* are accessible for such operations.

In (12), the inner DP lies within the domain of the phase head R. While the CIP moves to the Edge (Spec,RP) via predicate inversion, the complement of R (the inner DP) remains in situ. Consequently, when T probes for a goal to satisfy the EPP feature, the inner DP is invisible. Therefore, extracting the inner DP violates the PIC, correctly predicting the ungrammaticality of the derivation. However, since the RP constitutes part of the extended nominal projection, it can satisfy the EPP. As the only accessible goal, the entire RP moves to Spec,TP, as shown in (13).

(12) **PIC-violating movement of inner DP of RP to Spec,TP:**



(13) **Movement of RP to Spec,TP:**



3 The Syntax of Distributivity

Distributivity occurs in a sentence when the predicate is distributed to an individual or some parts of the plurality. In Korean, a particle *-ssik* is a distributivity marker that attaches to a noun phrase, as in sentences (14) and (15). Its counterparts are English *each* and German *je* (Link, 1991).

- (14) ai-tul two myeong-ssik-i chaek-ul sa-ess-ta.
 child-PL two CL-DIST-NOM book-ACC buy-PST-DECL
 ‘The children, in groups of two, bought a book.’ ≈ ‘The children, in pairs, bought a book.’
- (15) ai-tul han myeong-ssik-i chaek-ul sa-ess-ta.
 child-PL one CL-DIST-NOM book-ACC buy-PST-DECL
 ‘The children, in groups of one, bought a book.’ ≈ ‘Each child bought a book.’

3.1 Properties of the Marker *-ssik*

The distributive marker *-ssik*, which is the head of DistP, creates a partition of the head noun. The cardinality of this partition is determined by the #P⁶ or CIP that the marker follows, as in (16). Syntactically, strict structural adjacency is required. That is, no element can surface between the CIP and the marker.

- | | |
|------------------------------------|------------------------------|
| (16) a. CIP adjoining DistP | b. #P adjoining DistP |
| two myeong-ssik | twul ssik |
| two CL _{human} -DIST | two DIST |
| ‘two (person) each’ | ‘two each’ |

It is important to note that the distribution of *-ssik* is not limited to nominal environments⁷. Unlike numeral classifiers, which are strictly adnominal, *-ssik* exhibits categorial flexibility, attaching to degree adverbs (e.g., *jogeu-ssik* ‘little by little’) or temporal adverbs (e.g., *gakkeum-ssik* ‘occasionally’). This flexibility suggests that the DistP projection may function as an event (adverbial) modifier in addition to its role as a nominal modifier. In (17), *-ssik* attaches to adverbs (*jogeu* and *gakkeum*) to form event modifiers.

- (17) **AdvP adjoining DistP**
- a. ai-tul-i jogeu-ssik chaek-ul sa-ess-ta.
 child-PL-NOM little-DIST book-ACC buy-PST-DECL
 ‘The children moved little by little.’
- b. ai-tul-i gakkeum-ssik chaek-ul sa-n-ta.
 child-PL-NOM occasionally-DIST book-ACC buy-NPST-DECL
 ‘The children come occasionally.’

Consequently, a sentence can have two DistP that targets different components:

- (18) **Use of both Nominal and Adverbial DistP:**
- ai-tul two myeong-ssik-i jogeu-ssik chaek-ul sa-ess-ta.
 child-PL two CL-DIST-NOM little-DIST book-ACC buy-PST-DECL
 ‘Children, in groups of two, bought a book little by little.’

⁶#P can merge with DistP only if the numeral is in base form. As the syntactic effects are analogous, this paper focuses on CIP.

⁷This paper focuses on the nominal DistP.

3.2 Distributive Constructions

The structural adjacency of CIP and DistP accounts for the availability of the three types of numeral-classifier constructions, as shown in (19)⁸.

- (19) ‘Children, in groups of two, bought a book.’
- a. **NI Type with DistP:**
ai-tul **twu myeong-ssik-i** chaek-ul sa-ess-ta.
child-PL **two CL-DIST-NOM** book-ACC buy-PST-DECL
 - b. **FQ Type with DistP:**
ai-tul-i **twu myeong-ssik(-i)** chaek-ul sa-ess-ta.
child-PL-NOM **two CL-DIST(-NOM)** book-ACC buy-PST-DECL
 - c. **GC Type with DistP:**
twu myeong-ssik-ui ai-tul-i chaek-ul sa-ess-ta.
two CL-DIST-GEN child-PL-NOM book-ACC buy-PST-DECL

Analogous to CIP stranding, DistP stranding is restricted to the FQ type constructions involving temporal or modal adverbs, as in (20).

- (20) ‘Children, in groups of two, bought a book.’
- a. **NI Type with DistP and Temporal Adverb:**
ai-tul **twu myeong-ssik-i** eojae chaek-ul sa-ess-ta.
child-PL **two CL-DIST-NOM** yesterday book-ACC buy-PST-DECL
 - b. **FQ Type with DistP and Temporal Adverb (DistP stranding):**
ai-tul-i eojae **twu myeong-ssik(-i)** chaek-ul sa-ess-ta.
child-PL-NOM yesterday **two CL-DIST(-NOM)** book-ACC buy-PST-DECL
 - c. **GC Type with DistP and Temporal Adverb:**
twu myeong-ssik-ui ai-tul-i eojae chaek-ul sa-ess-ta.
two CL-DIST-GEN child-PL-NOM yesterday book-ACC buy-PST-DECL

Note that the restriction observed in (5) that no adverbs can surface between the head DP and CIP in the NI and GC types extends to DistP constructions:

- (21) **No adverbs can surface between DP and DistP in NI and GC Types:**
- a. *ai-tul eojae **twu myeong-ssik-i** chaek-ul sa-ess-ta.
child-PL yesterday **two CL-DIST-NOM** book-ACC buy-PST-DECL
Intended: ‘Children, in groups of two, bought a book yesterday.’
 - b. ***twu myeong-ssik-ui** eojae ai-tul-i chaek-ul sa-ess-ta.
two CL-DIST-GEN yesterday child-PL-NOM book-ACC buy-PST-DECL
Intended: ‘Children, in groups of two, bought a book yesterday.’

3.3 Structural Proposal: DistP as Modifier

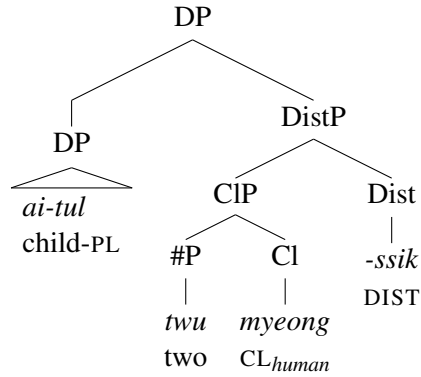
Given that the syntactic position of *-ssik* is strictly restricted to the position following CIP, I propose that *-ssik* is a functional head that selects CIP as its complement, as shown in (22). Furthermore, I analyze the

⁸DistPs are in bold.

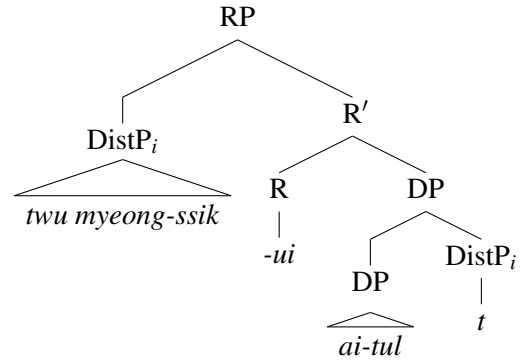
GC type (22b) as an instance of DP-internal predicate inversion involving DistP, a derivation analogous to that of CIP in (7b).

(22) **Internal Structure of DPs with DistP**

a. **NI and FQ Types:**



b. **GC Type:**

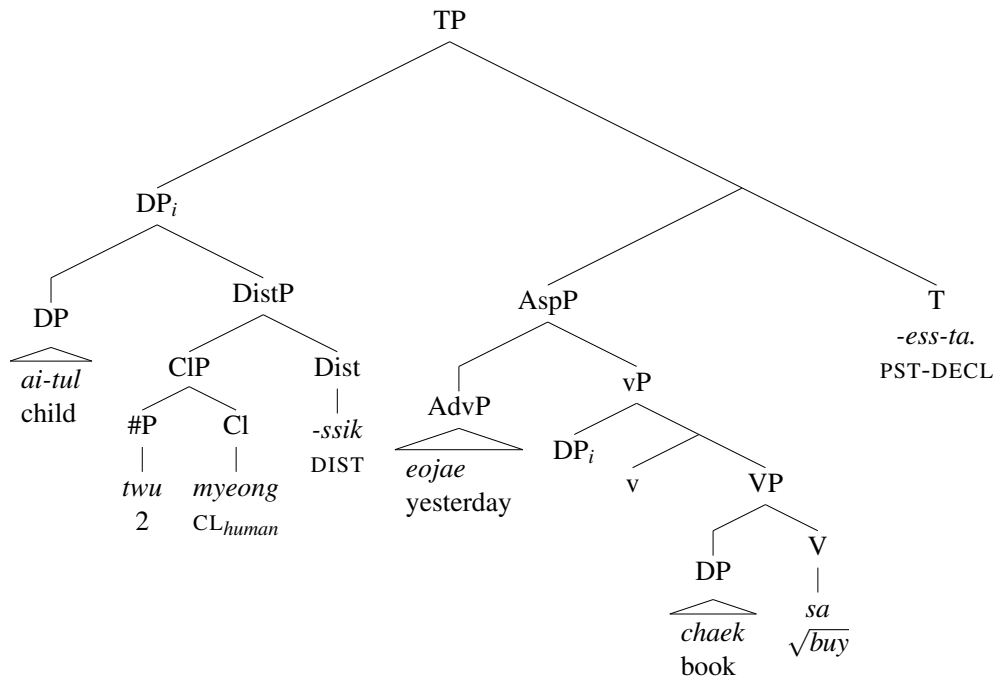


3.4 DistP Stranding

Similar to CIP stranding in Section 2.4, I propose that the grammaticality of (20b) is a result of DistP stranding. That is, in the base construction of the complex DP (22a), the movement of the complex DP to Spec,TP position produces the NI type (20a) while the movement of the inner DP produces the FQ type (20b). The GC type (20c) is a result of the movement of the complex DP with predicate inversion (22b).

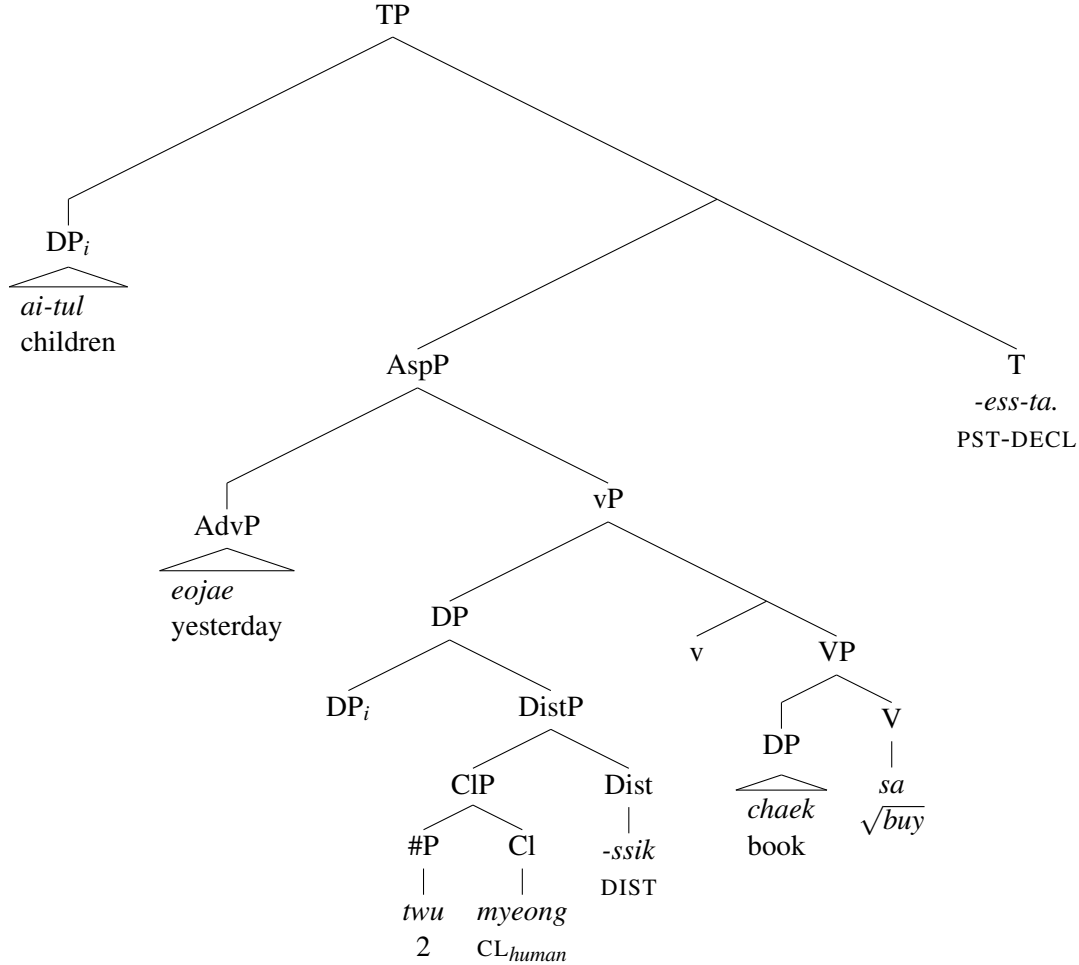
In (23), the movement of the entire complex DP *ai-tul twu myeong-ssik* to Spec,TP is made to satisfy the EPP feature of T. As DistP is a modifier of the head DP, it is possible to be pied-piped and surface before the temporal adverb *eojae*. Then, the case assignment happens. The complex DP *ai-tul twu myeong-ssik* receives NOM case and the lower DP *chaek* receives ACC case.

(23) **Movement of complex DP to Spec,TP:**



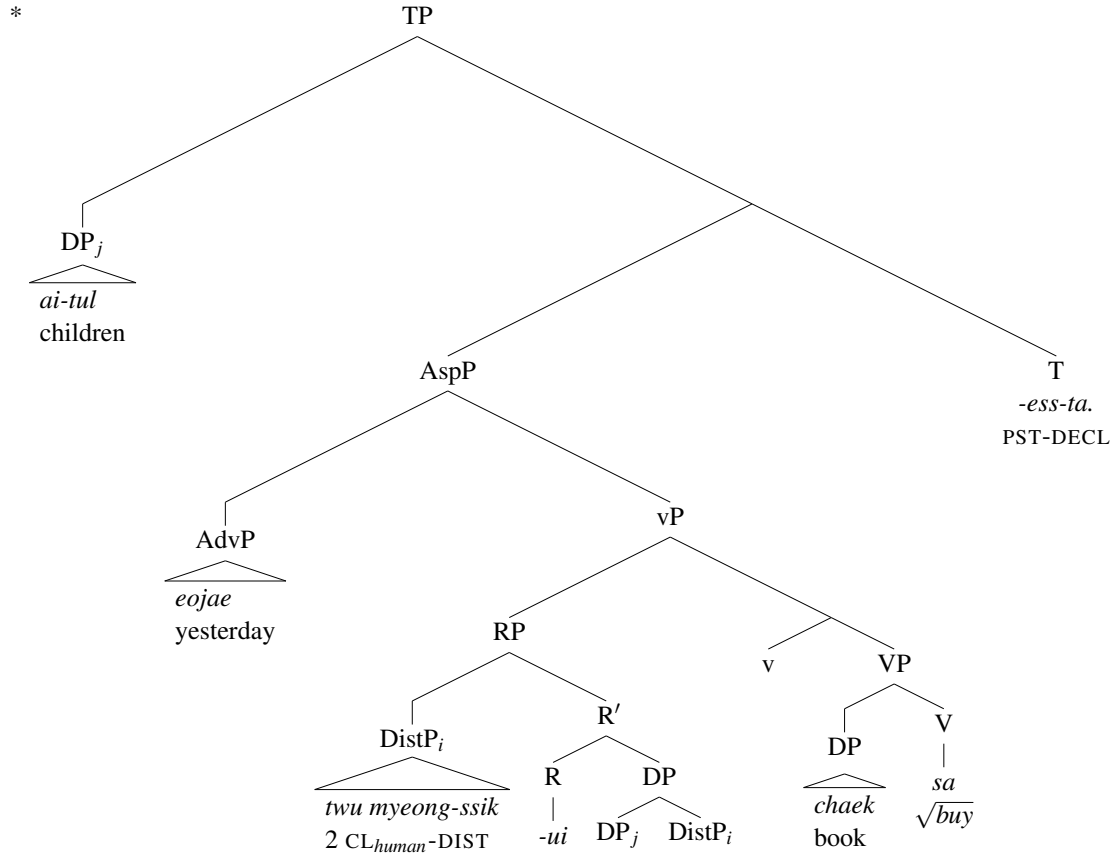
In (24), DistP stranding is derived via the same mechanism proposed for CIP stranding. Because the inner DP is in the phase edge, it remains visible to T. Consequently, the inner DP alone moves to Spec,TP to satisfy the EPP, stranding DistP in its base-generated position. After the movement, the inner DP *ai-ul* receives NOM case, the lower DP *chaek* receives ACC case and stranded DistP has an optional NOM marking.

(24) **Movement of the inner DP to Spec,TP (DistP stranding):**

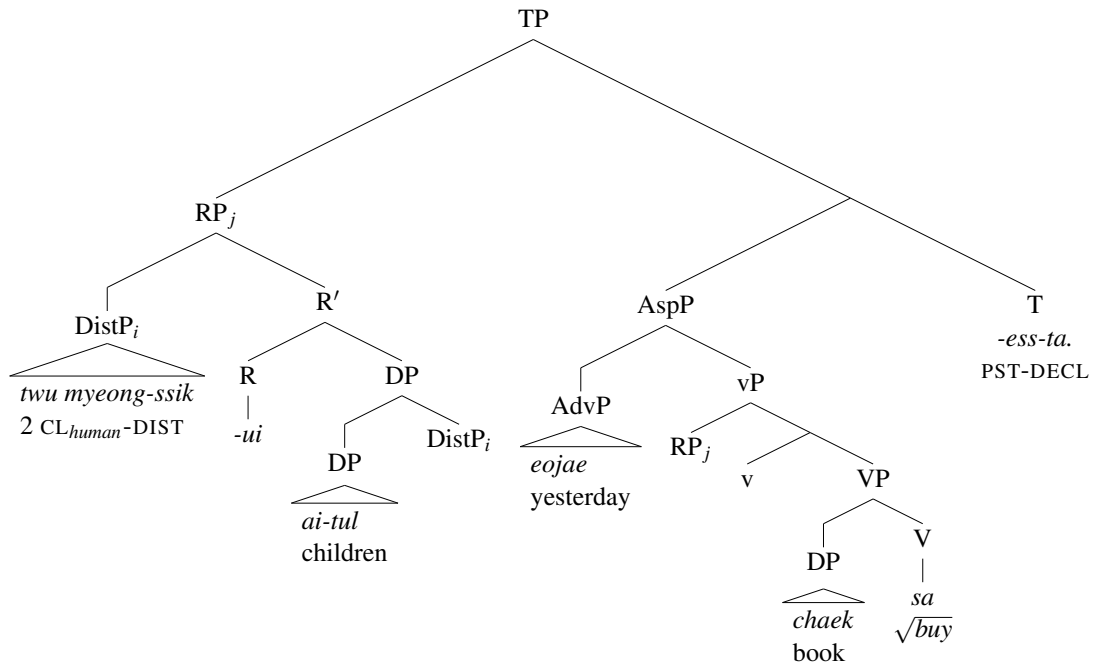


As expected from the observation of CIP stranding not possible under the GC type (Section 2.4.2), the movement of inner DP in the predicate-inverted complex DP *twu myeong-ssik-ui ai-tul* is not possible. That is, DistP stranding is not possible under the GC type, as in (25). The movement of the entire complex DP is possible, as in (26).

(25) **PIC-violating movement of inner DP of RP to Spec,TP:**



(26) **Movement of RP to Spec,TP is possible:**



4 Complex Distributive Constructions

4.1 Complex DistP

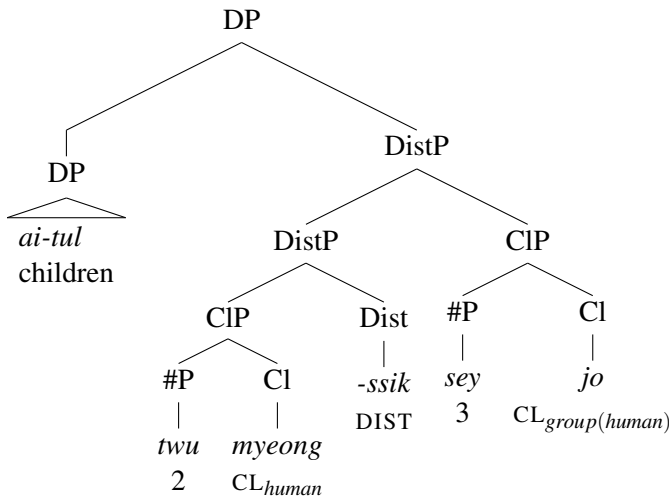
In addition to the CIP selected as the complement of the Dist head, a secondary CIP can be adjoined to the maximal projection as a modifier. I refer to this configuration as a **Complex DistP**. As discussed in Section 3.1, the presence of the lower CIP (the complement) is obligatory, whereas the higher CIP is optional. In (27a), the higher CIP functions as a counter for the group of children, evidenced by the use of the classifier *jo* ('group'). A parallel example involving non-human nouns is provided in (27b)⁹.

(27) **Complex DistP Constructions:**

- a. *ai-tul twu myeong-ssik sey jo*
 child-PL **two** CL_{human}-DIST **three** CL_{human_group}
 ≈ 'children, in three groups of two' (total 6 children)
- b. *chaek-tul sey kwon-ssik daseot mukkeum*
 book-PL **three** CL_{book}-DIST **five** CL_{paper_group}
 ≈ 'books, in five bundles of three' (total 15 books)

The structural configuration of (27a), showing the adjunction of the higher CIP, is shown in (28).

(28) **Syntactic Structure of (27a):**



The structural parallelism between the bare CIP and the Complex DistP is evident in their movement patterns. In the NI type (29a), the entire complex DP, which contains the head noun and the adjoined Complex DistP, undergoes pied-piping to Spec,TP. In the FQ type (29b), only the head DP *ai-tul* undergoes movement, resulting in the stranding of the Complex DistP in its base-generated position. The GC type is also possible, as shown in (29c); here, the entire Complex DistP undergoes Predicate Inversion to Spec,RP, followed by the movement of the entire RP to Spec,TP.

(29) ≈ 'Children, in three groups of two, bought a book yesterday.'

a. **NI Type with DP, Complex DistP and Temporal Adverb:**

- ai-tul twu myeong-ssik sey jo-ga eojae chaek-ul sa-ess-ta.*
 child-PL **two** CL-DIST **three** CL-NOM yesterday book-ACC buy-PST-DECL

⁹Complex DisPs are in bold.

b. **FQ Type with DP, Complex DistP and Temporal Adverb:**

ai-tul-i eojae **twu myeong-ssik sey jo(-ga)** chaek-ul sa-ess-ta.
 child-PL-NOM yesterday **two CL-DIST three CL(-NOM)** book-ACC buy-PST-DECL

c. **GC Type with DP, Complex DistP and Temporal Adverb:**

twu myeong-ssik sey jo-ui ai-tul-i eojae chaek-ul sa-ess-ta.
two CL-DIST three CL-GEN child-PL-NOM yesterday book-ACC buy-PST-DECL

4.2 Independent Adjunction of DistP and CIP

Since CIPs function as modifiers, they can merge directly to the head DP, irrespective of the presence of a DistP. Thus, the head DP can be modified by both a bare CIP and a DistP independently, appearing in various surface orders as shown in (30a) and (30b).

(30) ≈ ‘twenty children, in groups of two’ (total 20 children)

a. **DistP adjoining to [DP + CIP]:**

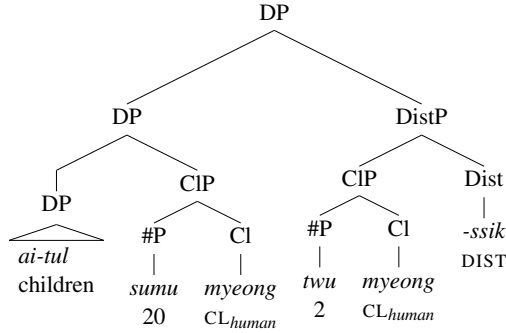
ai-tul sumu myeong twu myeong-ssik
 child-PL twenty CL two CL-DIST

b. **CIP adjoining to [DP + DistP]:**

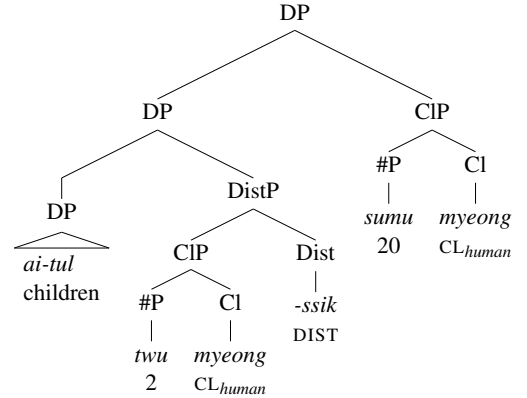
ai-tul twu myeong-ssik sumu myeong
 child-PL two CL-DIST twenty CL

Structure configurations of (30), showing possible orders of adjunction, are shown in (31).

(31) a. **Syntactic Structure of (30a):**



b. **Syntactic Structure of (30b):**



4.3 Complex DistP Stranding

Analogous to CIP and DistP stranding, Complex DistP stranding is restricted to the FQ type constructions involving temporal or modal adverbs, as in (32).

(32) ‘Children, in three groups of two, bought a book.’

a. **NI Type with Complex DistP and Temporal Adverb:**

ai-tul **twu myeong-ssik sey jo-ga** eojae chaek-ul sa-ess-ta.
 child-PL **two CL-DIST three CL-NOM** yesterday book-ACC buy-PST-DECL

b. **FQ Type with Complex DistP and Temporal Adverb (Complex DistP stranding):**

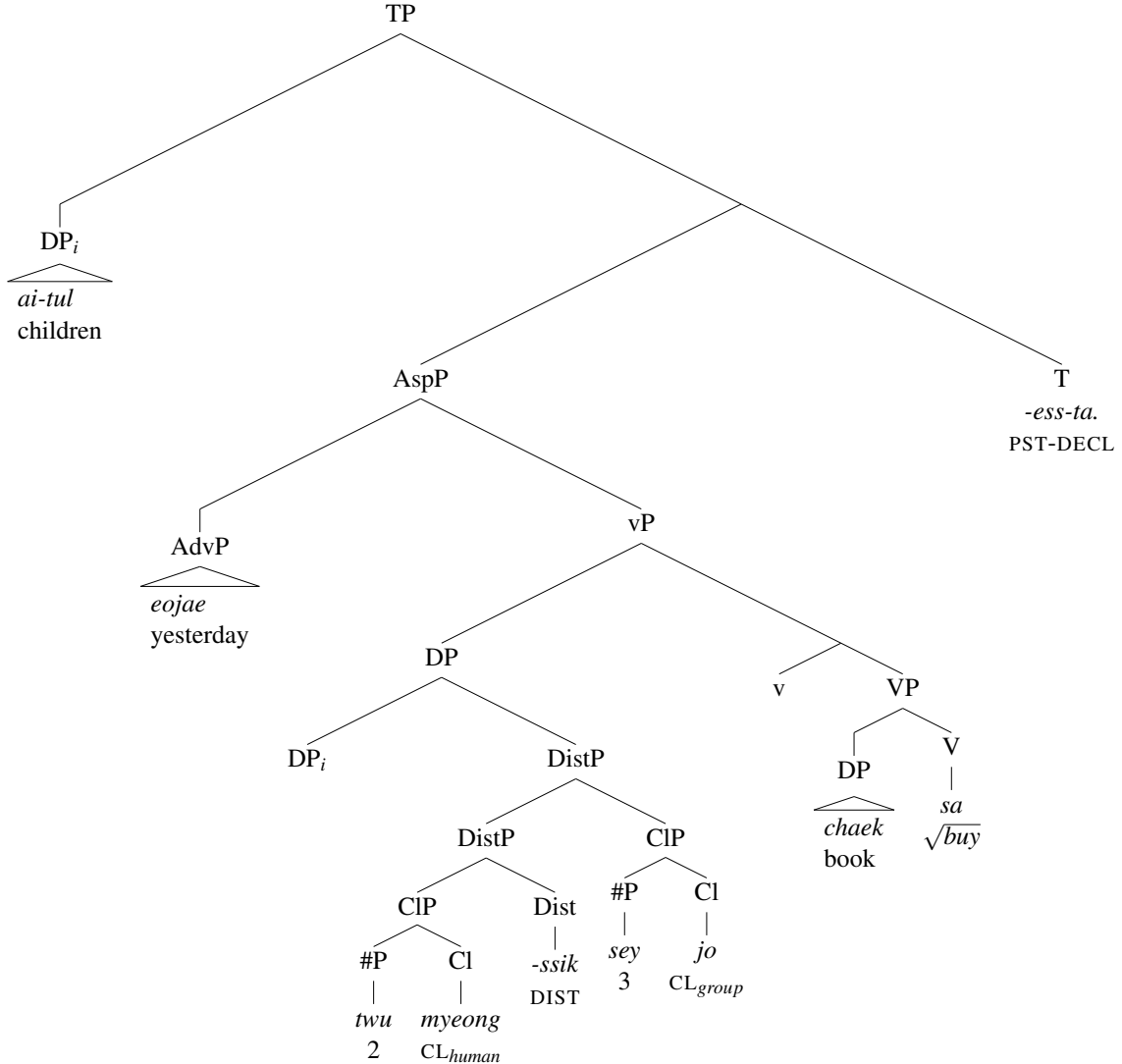
ai-tul-i eojae **twu myeong-ssik sey jo(-ga)** chaek-ul sa-ess-ta.
 child-PL-NOM yesterday **two CL-DIST** **three CL(-NOM)** book-ACC buy-PST-DECL

c. **GC Type with Complex DistP and Temporal Adverb:**

twu myeong-ssik sey jo-ui ai-tul-i eojae chaek-ul sa-ess-ta.
two CL-DIST **three CL-GEN** child-PL-NOM yesterday book-ACC buy-PST-DECL

The syntactic structure of (32b), which is an evidence to the movement of inner DP of the complex DP and stranding complex DistP in situ, is shown in (33).

(33) **Movement of inner DP to Spec,TP (Complex DistP stranding):**



4.4 Stranding in Complex DPs with additional DistP or ClP

In ClP, DistP and Complex DP stranding, it is observed that the stranding happens only in the FQ type so this section focuses on the FQ type only. Unlike those stranding, a complex DP with independently adjoined DistP and ClP has two modifiers. That is, while complex DPs with single modifiers do not have an optionality of which phrase to strand (as there is only one phrase), complex DPs with two modifiers have such optionality.

4.4.1 Stranding in [[DP + DistP] + CIP]

In the FQ type with a complex DP structure of [[DP + DistP] + CIP], all three movements are possible: (34a) the maximal projection, (34b) the intermediate DP (stranding the outer CIP), or (34c) the head DP (stranding both DistP and CIP).

- (34) ≈ ‘Twenty children, in groups of two, bought a book yesterday.’
- a. **Maximal Pied-Piping (Movement of [[DP + DistP] + CIP]):**
 ai-tul **twu myeong-ssik sumu** myeong-i eojae chaek-ul sa-ess-ta.
 child-PL **two CL-DIST** **twenty** CL-NOM yesterday book-ACC buy-PST-DECL
 - b. **Intermediate Stranding (Movement of [DP + DistP]):**
 ai-tul **twu myeong-ssik-i** eojae **sumu** myeong(-i) chaek-ul sa-ess-ta.
 child-PL **two CL-DIST-NOM** yesterday **twenty** CL(-NOM) book-ACC buy-PST-DECL
 - c. **Deepest Stranding (Movement of DP):**
 ai-tul-i eojae **twu myeong-ssik sumu** myeong-i chaek-ul sa-ess-ta.
 child-PL-NOM yesterday **two CL-DIST** **twenty** CL(-NOM) book-ACC buy-PST-DECL

4.4.2 Stranding in [[DP + CIP] + DistP]

However, in the FQ type with a complex DP structure of [[DP + CIP] + DistP], not all movements are possible. While the intermediate DP (35b) and the head DP (35c) movement to Spec,TP is still possible, the movement of entire complex DP (maximal pied-piping; 35a) is not possible.

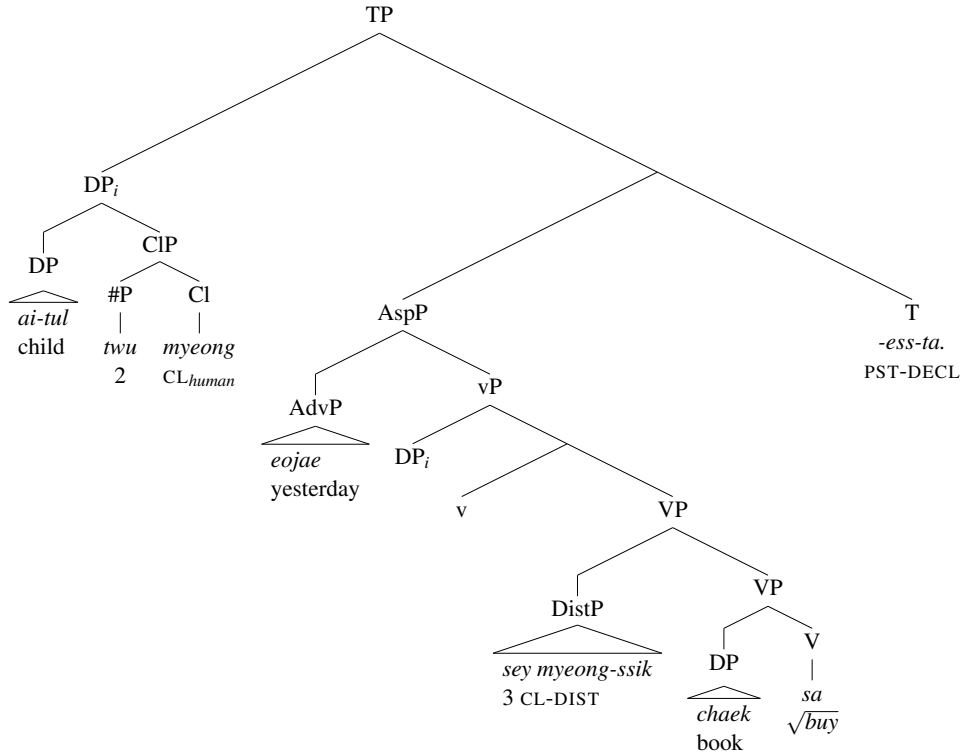
- (35) ≈ ‘Twenty children, in groups of two, bought a book yesterday.’
- a. **Blocked Maximal Pied-Piping (Movement of [[DP + CIP] + DistP]):**
 *ai-tul **sumu myeong twu myeong-ssik-i** eojae chaek-ul sa-ess-ta.
 child-PL **twenty CL** **two** CL-DIST-NOM yesterday book-ACC buy-PST-DECL
 - b. **Intermediate Stranding (Movement of [DP + CIP]):**
 ai-tul **sumu myeong-i** eojae **twu myeong-ssik(-i)** chaek-ul sa-ess-ta.
 child-PL **twenty CL-NOM** yesterday **two** CL-DIST(-NOM) book-ACC buy-PST-DECL
 - c. **Deepest Stranding (Movement of DP):**
 ai-tul-i eojae **sumu myeong twu myeong-ssik(-i)** chaek-ul sa-ess-ta.
 child-PL-NOM yesterday **twenty CL** **two** CL-DIST(-NOM) book-ACC buy-PST-DECL

4.4.3 Constraints in DistP adjoining to DP

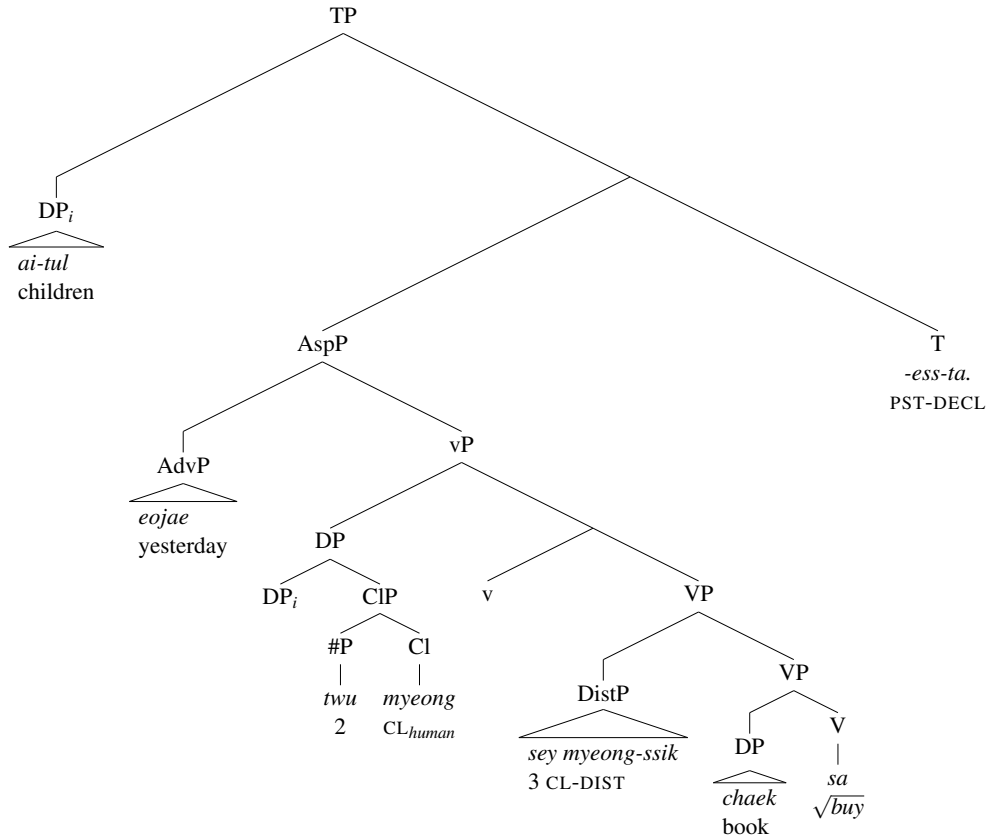
The asymmetry in maximal pied-piping of DPs with both DistP and CIP, as in (34a) and (35a), is unexpected as the movement of the entire DP was always possible in all stranding cases. As an explanation to this observation, I claim that a construction of [[DP + CIP] + DistP] is not possible. That is, considering that DistP can function as an event modifier (as mentioned in Section 3.1) and the agreement between N and CL is not strictly syntactic, (35a) is ungrammatical as the DistP *twu myeong-ssik* is an event modifier that is adjoined to VP. This analysis then explains the grammaticality of (35b) and (35c) by considering those as an instance of a simple CIP stranding. In (36) and (37), as DistP is not part of the nominal projection, it is invalid for movement along with the complex DP.

Then what accounts for the blocking of DistP adjoining to the complex DP with CIP? I propose that it is due to the ϕ -feature mismatch. DistP is [-cardinality] while CIP is [+cardinality] as it always carries a numeric value which marks the cardinality of the head noun. This allows DistP merging with bare DP as it is [-cardinality] while blocking DistP merging with complex DP with [+cardinality].

(36) Movement of complex DP to Spec,TP with DistP modifying VP; (35b):



(37) Movement of inner DP to Spec,TP with DistP modifying VP; (35c):



5 Conclusion

This paper has investigated the syntactic structure of numeral classifier constructions in Korean, focusing on the derivation of the three primary surface patterns: Noun Initial (NI), Floated Quantifier (FQ), and Genitive Case (GC). In contrast to previous analyses that treat the classifier as a functional head selecting the noun, I proposed that the CIP functions as a modifier to the head DP.

Under this modifier analysis, the variation in surface word order is explained by stranding. I argued that the NI type is a result of the movement of the maximal projection (pied-piping), while the FQ type results from the extraction of the inner DP, stranding CIP in its base-generated position. This analysis is further supported by the ungrammaticality of stranding in the GC construction. By identifying the GC type as an instance of predicate inversion within a Relator Phrase (RP), I showed that the inner DP is trapped within the domain of the RP phase. Thus, the Phase Impenetrability Condition (PIC) correctly predicts that only the entire RP is accessible for movement, blocking the derivation of a floated quantifier.

This structural analysis was extended to the distributivity marker *-ssik*. I proposed that while *-ssik* heads a DistP taking CIP as a complement, the maximal DistP projection functions as a modifier to the head DP. Consequently, DistP exhibits the same distributional properties as a bare CIP, allowing for both pied-piping and stranding.

Finally, the examination of complex distributive constructions revealed a constraint on DistP adjunction that it has a feature [-cardinality]. As CIP is [+cardinality] in that it provides specific cardinality of the noun, a ϕ -feature mismatch happens, preventing DistP merging with complex DP with CIP.

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